

## Trainings ▾

### Preparatory Steps

#### Rail Applications

##### **⚠ WARNING ⚠**

**When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.**

##### **⚠ WARNING ⚠**

**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

##### **⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**

Clean the area around the hose connections with steam.



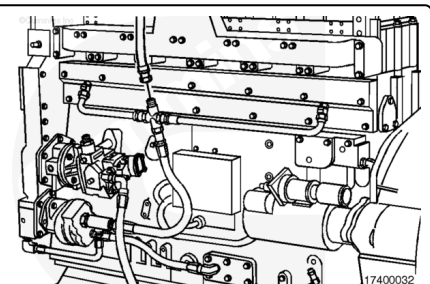
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### Remove

#### with Mechanically Actuated Injector

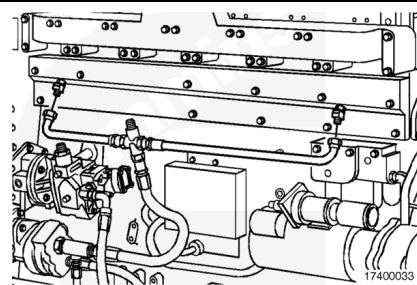
Disconnect the turbocharger oil drain hose from the cross.

Discard the face seal o-ring.



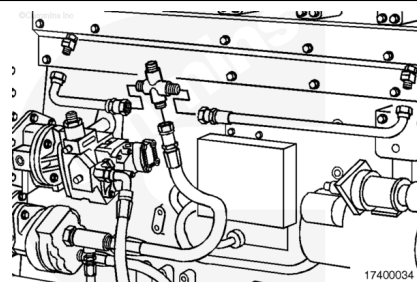
Disconnect the two scavenge tubes from the unions on the cam follower cover.

Discard the face seal o-rings.



Remove the cross from the cam follower cover tubes and the oil manifold supply hose.

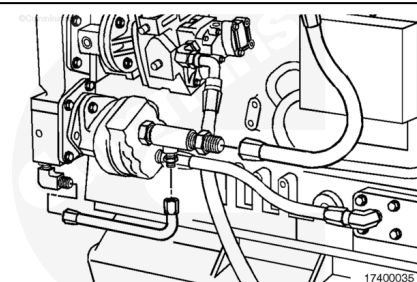
Discard the face seal o-rings.



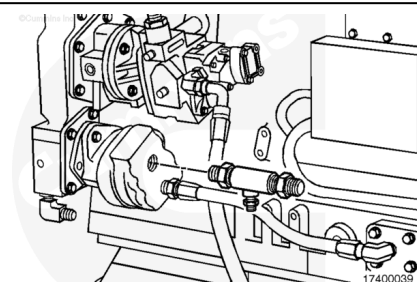
Remove the oil manifold supply hose from the oil inlet manifold connection.

Disconnect the hose from the elbow on the front gear cover and the oil inlet manifold connection.

Discard the face seal o-rings.

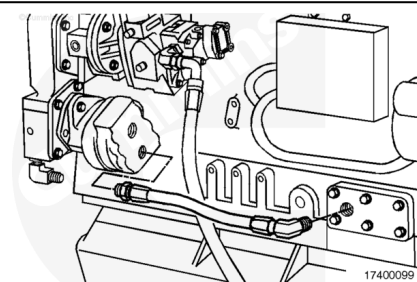


Disconnect the oil inlet manifold from the scavenge pump.

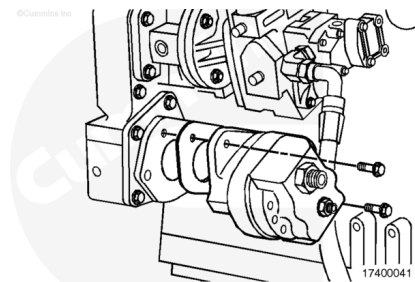


Disconnect the oil scavenge return hose from the scavenge pump fitting and the lubricating oil pan adapter hand hole cover.

Discard the face seal o-rings.



Remove the two capscrews, lubricating oil scavenge pump, and gasket from the hydraulic drive assembly. Discard the gasket.

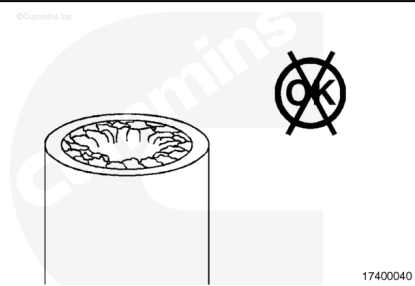


## Inspect for Reuse

### with Mechanically Actuated Injector

Check the lubricating oil scavenge pump lines for restrictions or wear.

Replace the hose if damage or restriction is found.

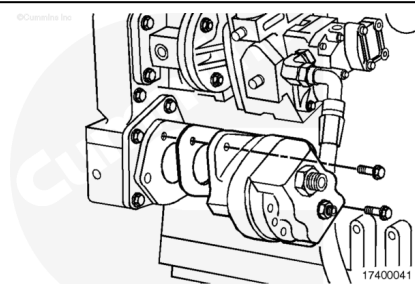


## Install

### with Mechanically Actuated Injector

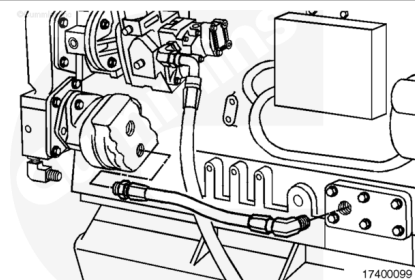
Install the gasket, lubricating oil scavenge pump, and two capscrews on the hydraulic drive assembly.

Tighten the capscrews.



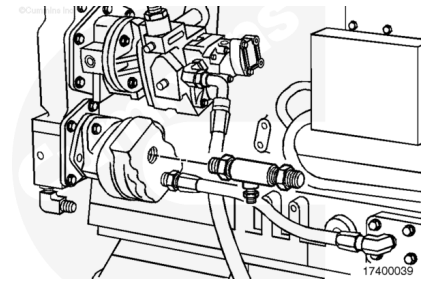
Install the face seal o-rings.

Connect the oil scavenge return hose to the lubricating oil pan adapter hand hole cover and the scavenge pump fitting.



Install the oil inlet manifold to the scavenge pump.

**Torque Value:** 95 n•m [ 70 ft-lb ]



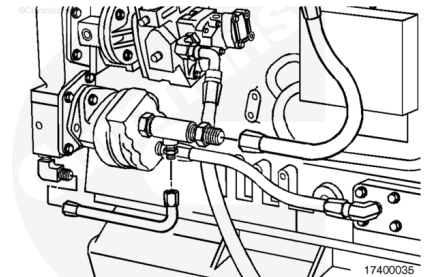
Install the face seal o-rings.

Connect the hose to the oil inlet manifold fitting and the front gear cover elbow.

**Torque Value:** 60 n•m [ 44 ft-lb ]

Connect the oil manifold supply hose to the oil manifold inlet connection.

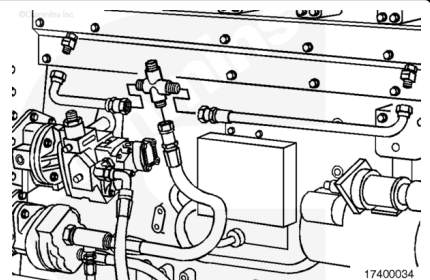
**Torque Value:** 80 n•m [ 59 ft-lb ]



Install the face seal o-rings.

Connect the cross to the oil manifold supply hose and the two scavenge tubes from the cam follower cover.

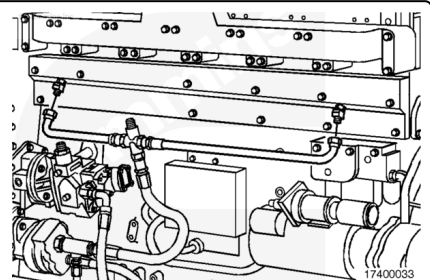
**Torque Value:** 80 n•m [ 59 ft-lb ]



Install the face seal o-rings.

Connect the two scavenge tubes to the front and rear of the cam cover.

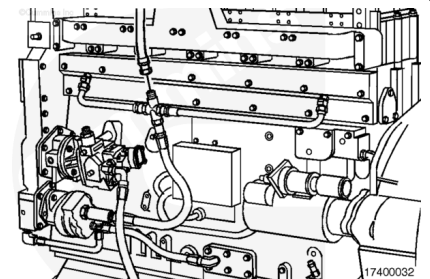
**Torque Value:** 80 n•m [ 59 ft-lb ]



Install the face seal o-rings.

Connect the turbocharger oil drain hose to the cross.

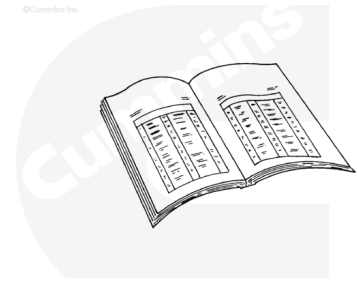
**Torque Value:** 80 n•m [ 59 ft-lb ]



## Finishing Steps

- Operate the engine and check for leaks.
- Check the oil level and add oil if required.

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**Last Modified: 30-Jun-2006**

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